

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 5-00-049

WASTE DISCHARGE REQUIREMENTS
FOR
HOUSING AUTHORITY OF THE COUNTY OF KERN
NORTH SHAFTER FARM LABOR CENTER NO. 5
KERN COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. The Kern County Housing Authority (hereafter Discharger) owns and operates North Shafter Farm Labor Center No. 5 (hereafter Center). Order No. 86-184, adopted by the Board on 26 September 1986, currently regulates the facility. The Discharger submitted a Report of Waste Discharge on 26 March 1999 because the on-site sewage disposal system is to be replaced as part of a camp reconstruction project. Additionally, Order No. 86-14 is neither adequate nor consistent with current plans and policies of the Board.
2. The Center is in the N $\frac{1}{2}$ of the NW $\frac{1}{4}$ of Section 5, T28S, R25E, MDB&M, as shown in Attachment A, which is part of this Order by reference.
3. The Center includes 108 residential units and two laundry facilities, which serve an estimated 475 residents about six months out of the year, approximately May to November.
4. Order No. 82-063, which regulated the site in the past, allowed for the use of an Imhoff tank followed by irrigation disposal on about 10 acres of land. The Imhoff tank was built in the 1940s when the site was used as an internment camp and has served its useful life. The tank was still being used at the time of a 30 September 1998 staff inspection, at which time it was noted that the tank was no longer adequate to treat domestic waste. There are also several septic tanks and leach lines currently in use at the camp.
5. The proposed method of treatment consists of septic tanks and leach fields at six locations around the Center. Four of the systems are for domestic waste and two of the systems are exclusively for laundry facilities. Most of the tanks and leachfields will be new, though four of the existing tanks and 759 feet of existing leachlines will remain. The entire site consists of more than 40 acres.
6. The proposed treatment and disposal systems are designed for a combined maximum flow of 70,525 gallons per day.
7. Percolation test boring logs submitted as part of the Report of Waste Discharge indicate that soils at the site consist of silty sand, sand, and sandy silt.

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8. Source water for the Center is supplied by an on-site well.
9. The California Department of Health Services has expressed concerns of high nitrate levels in the Center's drinking water. Septic systems may be a source of nitrates in groundwater, but so is irrigated agriculture. The supply well for the Center is upgradient from the existing and proposed disposal areas. The proposed system should not exacerbate the existing problem. Further, the drinking water goes through an ion exchange process to reduce nitrate concentrations to acceptable drinking water levels. The ion exchange regenerant waste is discharged to an offsite waste disposal facility permitted to accept the waste.
10. According to Mr. Alfred Rios, the Center's manager, groundwater depth is greater than 150 feet, and according to the Kern County Water Agency's 1995 *Water Supply Report*, depth to groundwater is approximately 250 feet.
11. The *Water Quality Control Plan for the Tulare Lake Basin, Second Edition*, (hereafter Basin Plan) designates beneficial uses, establishes water quality objectives, and contains implementation plans and policies for all waters of the Basin. These requirements implement the Basin Plan.
12. The site lies within the North Kern Hydrologic Area (No. 558.80) in the South Valley Floor Hydrologic Unit, as depicted on interagency hydrologic maps prepared by the California Department of Water Resources in August 1986.
13. Surface water drainage of the site is to the southwest. The nearest downgradient surface water is Jerry Slough, a Valley Floor Water, about nine miles west of the site. The beneficial uses of Valley Floor Waters, as identified in the Basin Plan, are agricultural supply; industrial service and process supply; water contact and noncontact water recreation; warm freshwater and wildlife habitat; rare and endangered species habitat; and groundwater recharge.
14. According to the Basin Plan, the beneficial uses of groundwater in the area are municipal, industrial, and agricultural supply.
15. The average annual rainfall in the area is 6.7 inches and the average annual pan evaporation rate is 61 inches.
16. The Board has considered antidegradation pursuant to State Water Resources Control Board Resolution No. 68-16 and finds that the permitted discharge is consistent with those provisions. The camp is used for six months, then closed during the winter. This allows the leach lines to rest each year and reduces the equivalent septic tank density to within accepted standards. The equivalent loading is unlikely to cause an increase in groundwater constituents above that of background levels. Land is available to replace or add additional leach lines if necessary.

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17. The action to update waste discharge requirements for this existing facility is exempt from the provisions of the California Environmental Quality Act (CEQA) in accordance with Title 14, CCR, Section 15301.
18. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
19. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Order No. 86-184 is rescinded and the Housing Authority of the County of Kern, its agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following at the North Shafter Farm Labor Center No. 5:

A. Discharge Prohibitions

1. Discharge of waste to surface waters or surface water drainage courses is prohibited.
2. Discharge of waste from the existing Imhoff tank (described in Finding No. 4) is prohibited.
3. Bypass or overflow of treated or untreated waste is prohibited.
4. Discharge of waste classified as 'hazardous,' as defined in Section 2521 (a) of Title 23, CCR, Section 2510, et seq., or 'designated,' as defined in Section 13173 of the California Water Code, is prohibited.

B. Discharge Specifications

1. The monthly average discharge shall not exceed 70,000 gallons per day.
2. Objectionable odors from the septic tank/leach field systems shall not be perceivable.
3. All settling and septic tanks shall be managed so as to prevent solids and grease carry-over into the leach fields. Specifically, the settling and septic tanks shall be pumped before sludge or grease will begin to carry-over into the leach fields, or when solids accumulation in the final tank is 75% of the usable solids storage volume of the tank, or the scum layer is within three inches of the bottom of outlet pipe, or the sludge level is within eight inches of the bottom of the outlet pipe, whichever is more restrictive.

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4. The maximum conductivity (hereafter EC) of the discharge shall not exceed the source water EC plus 500 $\mu\text{mhos/cm}$.
5. Septic tank and leach field locations shall comply with the following setback distances.

<u>Facility</u>	<u>Domestic Well</u>	<u>Public Well</u>	<u>Drainage Course¹</u>	<u>Cut or Fill Bank²</u>	<u>Property Line³</u>
Septic Tank	50 feet	100 feet	25 feet	10 feet	25 feet
Leach Field	100 feet	100 feet	50 feet	4h	50 feet

¹ As measured from the edge of the drainage course of stream.

² Distance in feet equals four times the vertical height of the cut or fill bank. Distance is measured from the top edge of the bank.

³ This distance shall be maintained when individual wells are to be installed and the minimum distance between waste disposal and wells cannot be assured.

C. Sludge Disposal Specifications

1. Collected sludges, scums, and other solids removed from liquid wastes shall be disposed of in a manner approved by the Executive Officer and consistent with *Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste*, as set forth in Title 27, CCR, Division 2, Subdivision 1, Section 20005, et seq.
2. Any proposed change in sludge use or disposal practice shall be reported to the Executive Officer and United States Environmental Protection Agency Regional Administrator **at least 90 days in advance** of the change.

D. Groundwater Limitations

The discharge, in combination with other sources, shall not cause underlying groundwater to contain waste constituents in concentrations statistically greater than background water quality, except EC. Regarding EC, the Discharger shall not cause underlying groundwater to exceed an incremental increase in EC greater than 25 $\mu\text{mhos/cm}$ over any five-year period.

E. Provisions

1. Septic tank cleanings shall be performed every six years, or more often as needed, and only by a duly authorized service.
2. The Discharger shall comply with Monitoring and Reporting Program No. 5-00-049, which is part of this Order, and any revisions thereto as ordered by the Executive Officer.

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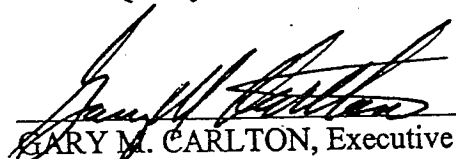
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3. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements," dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."
4. The Discharger must comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.
5. A copy of this Order shall be kept at the facility for reference by the personnel responsible for wastewater management. These persons shall be familiar with its contents.
6. In the event of any change in control or ownership of land or waste discharge facilities described herein, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.

To assume operation under this Order, the succeeding owner or operator must apply in writing to the Executive Officer requesting transfer of the Order. The request must contain the requesting entity's full legal name, the State of incorporation if a corporation, the name and address and telephone number of the persons responsible for contact with the Board, and a statement. The statement shall comply with the signatory paragraph of Standard Provision D.6 and state that the proposed owner or operator assumes full responsibility for compliance with this Order. Failure to submit the request shall be considered a discharge without requirements, a violation of the California Water Code and Clean Water Act. Transfer shall be approved or disapproved by the Executive Officer.

7. The Board will review this Order periodically and will revise requirements when necessary.
8. Authorization for 70,000 gpd is predicated on six months of use. Increased occupancy will be sufficient reason to cause reopening of this Order and more advanced wastewater treatment and control.

I, GARY M. CARLTON, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 17 March 2000.


GARY M. CARLTON, Executive Officer

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. 5-00-049

FOR
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Specific sample station locations shall be established with concurrence of Board staff, and a description of the stations shall be submitted to the Board and attached to this Program by 1 April 2000.

SEPTIC TANK MONITORING

All septic tanks shall be inspected and pumped as described below. An inspection is not required during the year a septic tank is pumped. Each septic tank shall be monitored as follows:

<u>Constituent</u>	<u>Units</u>	<u>Type of Measurement</u>	<u>Frequency</u>
Sludge Depth	feet	Staff Gauge	Annually ¹
Total Volume of Sludge (A)	ft ³	Calculated	Annually ¹
Total Usable Sludge Storage (B)	ft ³	Calculated	Annually ¹
Percent Full (A/B)	%	Calculated	Annually ¹
Scum Thickness	feet	Staff Gauge	Annually ¹
Distance Between Bottom of Scum Layer and Bottom of Outlet Device	inches	Staff Gauge	Annually ¹
Distance Between Top of Sludge Layer and Bottom of Outlet Device	inches	Staff Gauge	Annually ¹

¹ In lieu of septic tank measurement, the septic tanks may be pumped annually

SEPTIC TANK RECORDS

The Discharger shall maintain a log of all septic tank cleanings. At a minimum, the log shall include the date of the cleaning, and the name, address, phone number, and license number of the cleaner.

SEPTIC TANK EFFLUENT MONITORING

Septic tank effluent shall be monitored as follows:

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<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Frequency</u>
Flow ¹	gallons/day	Estimate	Monthly
Total Nitrogen	mg/l	Grab	Semi-Annually ²
EC	µmhos/cm	Grab	Semi-Annually ²

¹ Flow estimates shall be made by a method approved by the Executive Officer

² June, September

WATER SUPPLY MONITORING

A sampling station shall be established where a representative sample of the water supply can be obtained. Water supply monitoring shall include at least the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Frequency¹</u>
EC	µmhos/cm	Grab	Semi-Annually ²
Nitrate	mg/l	Grab	Semi-Annually ²

¹ Concurrent with effluent monitoring

² June, September

REPORTING

Monthly and annual monitoring shall be summarized in an annual report and submitted by 1 February of each year.

In reporting the monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in such a manner that illustrates clearly whether the Discharger complies with waste discharge requirements.

If the Discharger monitors any pollutant at the locations designated herein more frequently than is required by this Order, the results of such monitoring shall be included in the discharge monitoring report.

The Discharger may also be requested to submit an annual report to the Board with tabular and graphical summaries of the monitoring data obtained during the previous year. Any such request shall be made in writing. The report shall discuss any corrective actions taken or planned which may be needed to bring the discharge into full compliance with the waste discharge requirements.

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By 15 February of each year, the Discharger shall submit a written report to the Executive Officer containing the following:

- a. The names and general responsibilities of all persons in charge of wastewater treatment and disposal.
- b. The names and telephone numbers of persons to contact regarding wastewater disposal for emergency and routine situations.

All reports submitted in response to this Order shall comply with the signatory requirements in Standard Provision B.3.

The Discharger shall implement the above monitoring program on the first day of the month following adoption of this Order.

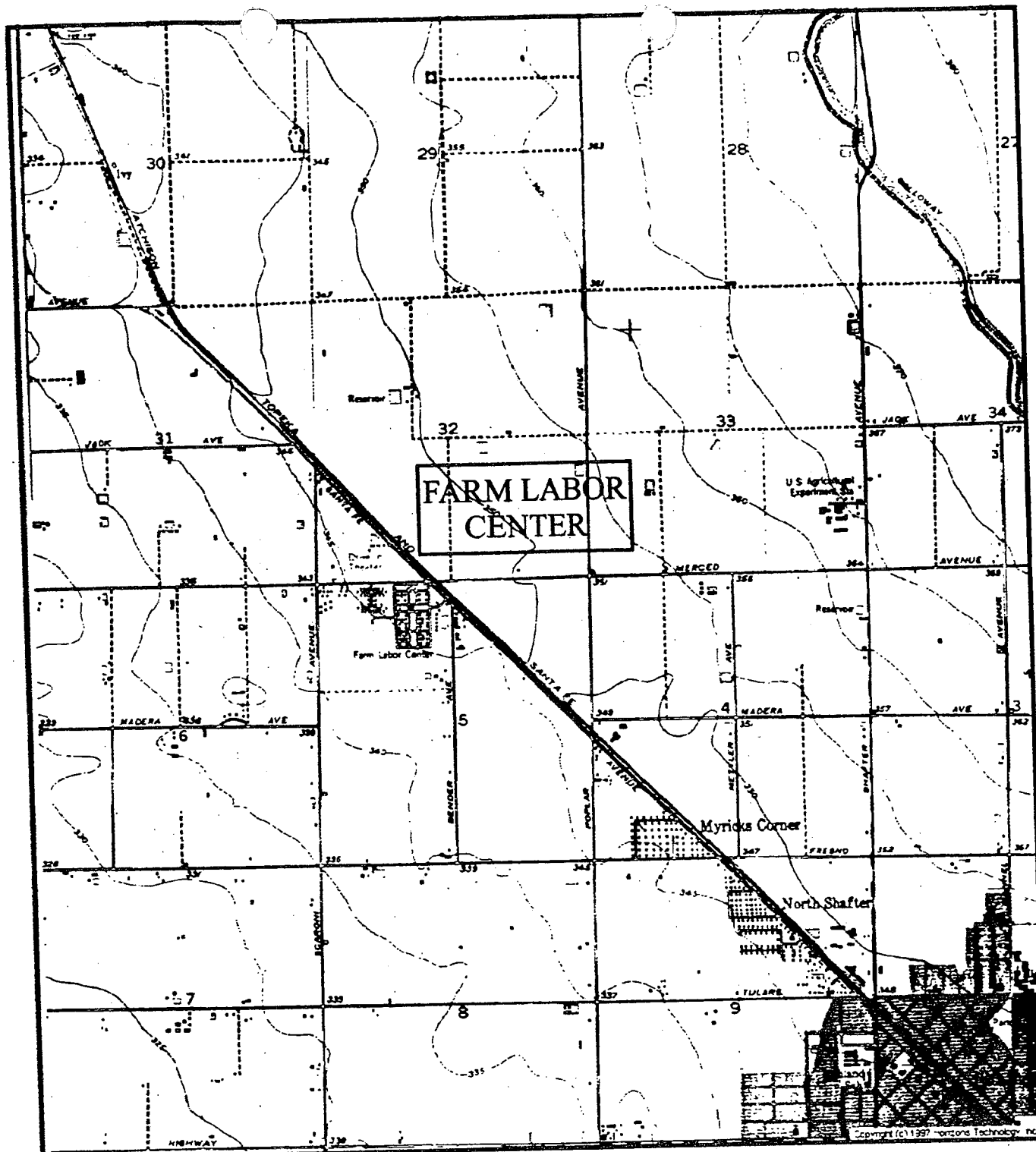
Ordered by:


GARY M. CARLTON, Executive Officer

17 March 2000

(Date)

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ATTACHMENT A

Vicinity Map

Order No. 5-00-049

HOUSING AUTHORITY OF THE COUNTY OF KERN

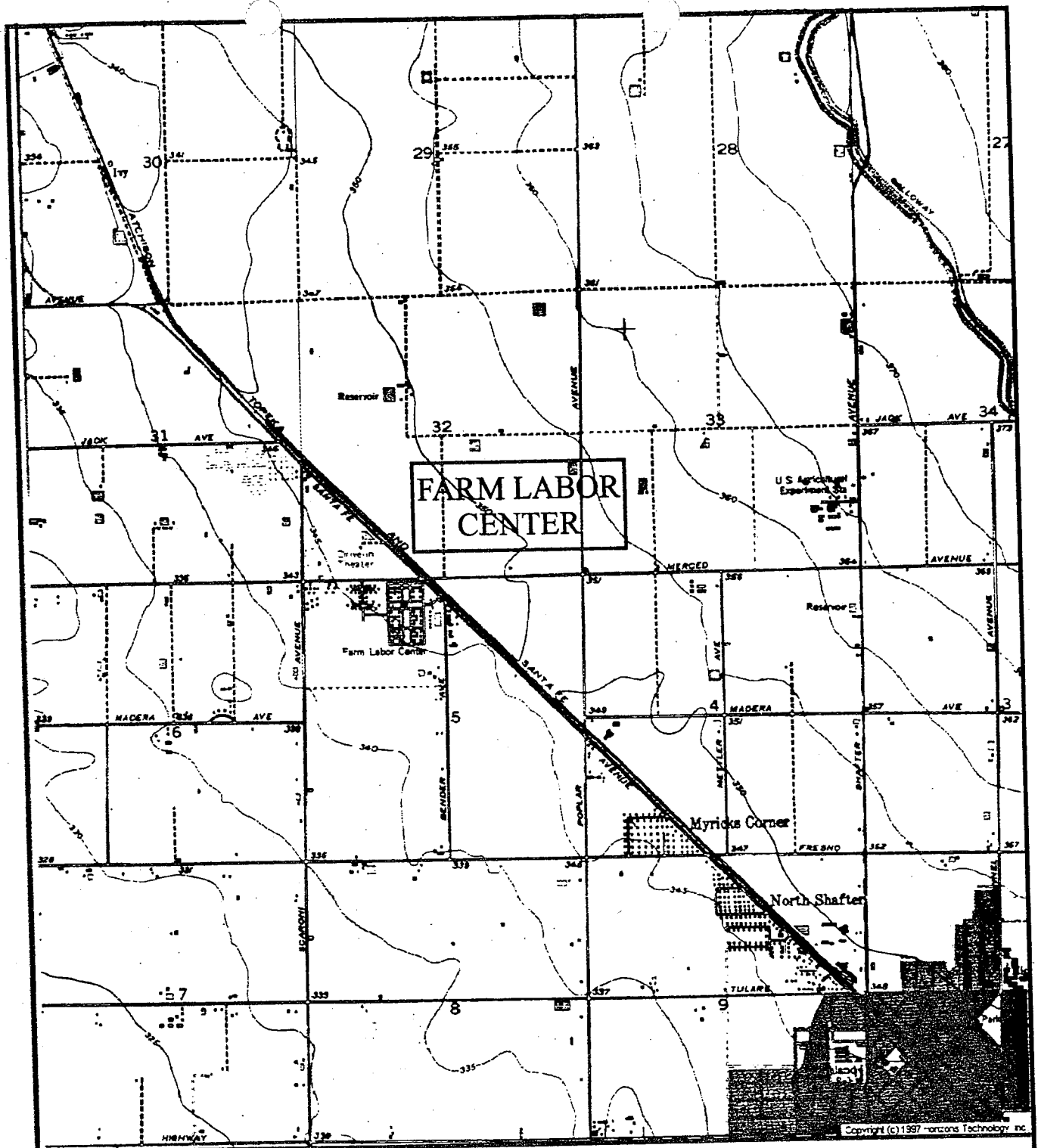
NORTH SHAFTER FARM LABOR CENTER

Section 5, T28S, R25E, MDB&M

Wasco, CA 7.5' USGS Quadrangles



1.2 IN = 1 KM



ATTACHMENT A

Vicinity Map
Order No.

HOUSING AUTHORITY OF THE COUNTY OF KERN

NORTH SHAFTER FARM LABOR CENTER

Section 5, T28S, R25E, MDB&M

Wasco, CA 7.5' USGS Quadrangles



1.2 IN = 1 KM

INFORMATION SHEET

ORDER NO. 5-00-049
HOUSING AUTHORITY OF THE COUNTY OF KERN
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KERN COUNTY

The Housing Authority of the County of Kern owns and operates North Shafter Farm Labor Center No. 5. The Center is at 17213 Central Valley Highway, approximately three miles northwest of Shafter.

The Center consists of 108 housing units, an office, a day care center, and two laundry facilities. Almost all of the Camp's residents (an estimated 475 people) stay for only six months of the year, approximately May to November.

Septic tanks and leach field, as well as an Imhoff tank, have been used for wastewater disposal in the past. Although some of the existing septic tanks and leach fields will be used in the proposed system, most will be removed. The Imhoff tank, no longer adequate to treat domestic waste, is prohibited from use by this Order.

The new facility is designed to treat and dispose of a maximum wastewater flow of 70,525 gallons per day. Like the old system, the new facility will direct domestic wastewater to one of four septic tank/leach field systems that are on the west and south sides of the Center's perimeter. Laundry wastewater will be directed to two separate septic tank/leach field systems in the middle of the Center. The following table summarizes the proposed system.

<u>System</u>	<u>No. of Tanks</u>	<u>Size of Tanks (gal/tank)</u>	<u>No. of Leach Lines</u>	<u>Length of Leach Lines (ft)</u>
1	4	5000	7	85
	1	1000		
2	1	6000	6	69
	1	4000	6	30
3	3	5000	3	52
	1	3000	3	50
			3	100
4	3	5000	5	100
	1	2000		
Laundry 1	1	3000	3	50
Laundry 2	1	6000	1	50
			3	63

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This Order includes groundwater limitations and septic tank/leach field system specifications contained in the *Water Quality Control Plan for the Tulare Lake Basin, Second Edition* (Basin Plan).

This Order requires the Discharger to either pump the septic tanks at least annually or inspect them at least annually to make sure that the septic tanks operate effectively and do not overflow. To assure that septage pumped from septic tanks is properly handled and disposed of at an authorized place for effective treatment, the Order requires the Discharger to use only licensed septic tank cleaning services.

The Order's effluent limitation for EC (500 μ mhos/cm over source water EC) is prescribed by the Basin Plan. The Center's source water is high in nitrate and must be treated to remove nitrate to below the maximum contaminant level of 10 mg/l as nitrogen. Domestic wastes contain nitrogen. Nitrogen in nitrate form is readily leachable and therefore, has the potential to affect groundwater. However, there should not be any impacts to groundwater at this site if the waste characteristics are consistent with typical domestic waste, and loading conforms to the seasonal occupancy described. With a forty-acre site and half-year use the equivalent septic system density conforms to accepted practice prescribed to preclude nitrate problems. This Order requires nitrogen monitoring to confirm this is the case.

The Basin Plan requires percolation rates to be faster than 60 minutes per inch but slower than five minutes per inch. If rates are faster than five minutes per inch, it must be shown that a sufficient distance of soil is available to assure proper filtration. Percolation test results show a range of 1.5 to 4.5 minutes per inch at the site; however, boring logs show that soils in the leach field areas consist of sand, silty sand, and sandy silt and depth to groundwater is over 150 feet, providing sufficient distance for proper filtration.

This is an existing facility and the action to update requirements is exempt from the provisions of the California Environmental Quality Act (CEQA) in accordance with Title 14, CCR, Section 15301.

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